

Rishabh Pandey

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SUMMARY

Data science and ML engineer skilled in **exploratory data analysis**, **statistical modeling**, and **predictive pipelines** using **Python**, **SQL**, and **PyTorch**. Combines strong **DSA foundations** with **NLP**, **LLM fine-tuning**, and **automated MLOps** to build scalable, production-ready backend systems.

EDUCATION

Vellore Institute of Technology

Bhopal, Madhya Pradesh

B.Tech in Computer Science and Engineering (AIML) – CGPA: 8.95/10

2023 – 2027

WORK EXPERIENCE

AI Intern

AMASQIS.AI, Remote

April 2025 – September 2025

- Fine-tuned **DeepSeek-R1-Distill-Llama-8B** for financial instruction tasks using **4-bit QLoRA (r=16)** with 7 projection modules, training for 3 epochs with a **2,048-token sequence length**.
- Developed an automated data pipeline to convert **10K+ multi-attribute customer records** into instruction-tuning data using a **70:20:10 stratified split**. Built an **XGBoost baseline** to generate eligibility probabilities across **5 core banking product categories**.
- Integrated the fine-tuned model with a **FastAPI** backend, adding **regex-based guardrails**, JSON formatting, schema validation, input error handling, and fallback logic to manage model outputs and ensure reliable responses.

PROJECTS

StorageIQ [Live](#) | Python, PyTorch, Scikit-learn, FastAPI, Docker, GitHub Actions

- Calculated **90-day hardware replacement risk (VaR₉₅ / CVaR₉₅)** across **107.9M+ drive-days (~312K production drives)** to optimize SRE capital budgeting, using a **10,000-path Gamma-Poisson Monte Carlo simulation** driven by an inverse-MAE weighted **PyTorch LSTM + ARIMA ensemble**.
- Engineered a leakage-free validation and anomaly detection framework pairing **Isolation Forests** with **3 σ Z-scores** on SMART health telemetry (SMART 5, 194, 197), demonstrating that the ensemble outperformed Naïve, Seasonal Naïve, and Damped ETS baselines with **MASE < 1** across **88 drive models**.
- Architected a **FastAPI** analytics backend with automated schema validation, monthly model retraining, and **CI/CD deployment using GitHub Actions**, eliminating runtime inference latency.

AtlasML [Live](#) | FastAPI, PostgreSQL, Scikit-learn, XGBoost, Pydantic, SQLAlchemy, MLflow, SHAP, Docker

- Improved training speed by **60%+** by removing unnecessary SARIMA calculations, reducing model iterations, and reusing processed data.
- Built an **automated ML tool** with a **human-in-the-loop** workflow that takes CSV datasets, checks the target column and dataset type, and lets users choose from **Supervised**, **Unsupervised**, or **Time-Series** models.
- Used **MLflow** to track model experiments and **SHAP** to understand feature importance, while using **Joblib** for model storage and **Docker Compose**, **Render**, and **Neon PostgreSQL** for deployment.

TECHNICAL SKILLS

Languages: Python, C++, SQL

Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Plotly

ML/DL Frameworks: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, CatBoost, Statsmodels

NLP & GenAI: Natural Language Processing, LLM Fine-Tuning, Hugging Face Transformers, Text Embeddings, SpaCy

Backend & Databases: FastAPI, Pydantic, SQLAlchemy, REST APIs, PostgreSQL, MySQL

MLOps & Tools: MLflow, SHAP, Docker, Git, GitHub, GitHub Actions

Coursework: Data Structures & Algorithms, OOPs, Computer Networks, Linear Algebra, Probability & Statistics

ACHIEVEMENTS

- Competitive Programming:** Achieved a **3-Star rating** on CodeChef and a **1,413 contest rating** on LeetCode, solving **350+ Data Structures & Algorithms problems** across platforms.
- Kaggle Competitions:** Secured a global rank of **1,232** in Kaggle data science competitions through predictive machine learning pipelines and statistical feature engineering.